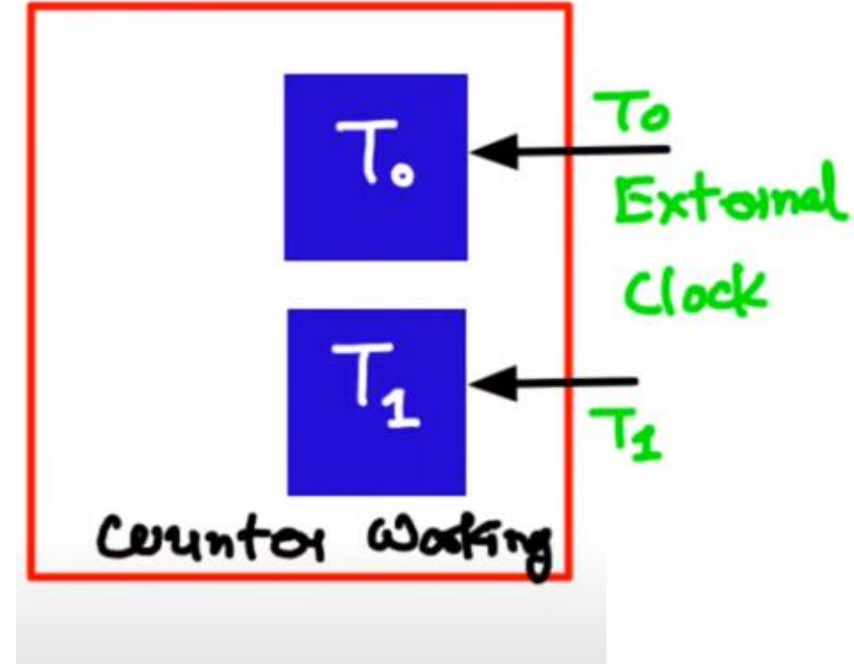
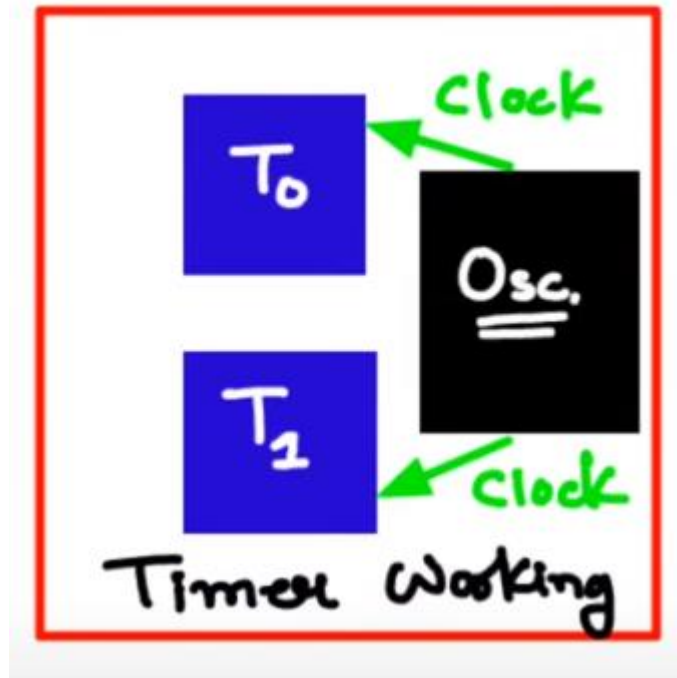


Lecture 5

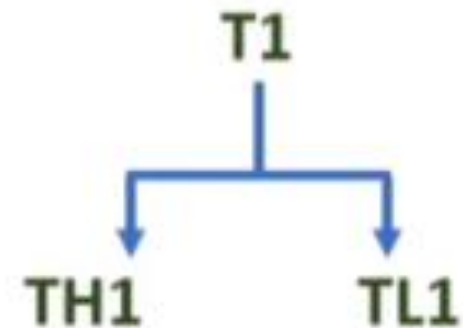
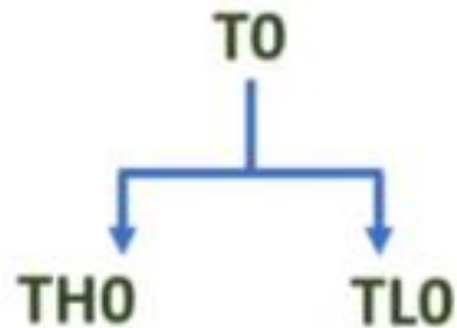
Timer and Counter

Timer and Counter of 8051



Timer and Counter of 8051

- ❑ 8051 has Two 16 bits Timers T0 & T1, working as up counters. T0 and T1 is further divided into 8bits of registers TH0-TL0 and TH1-TL1.



❑ **How to Load Count?**

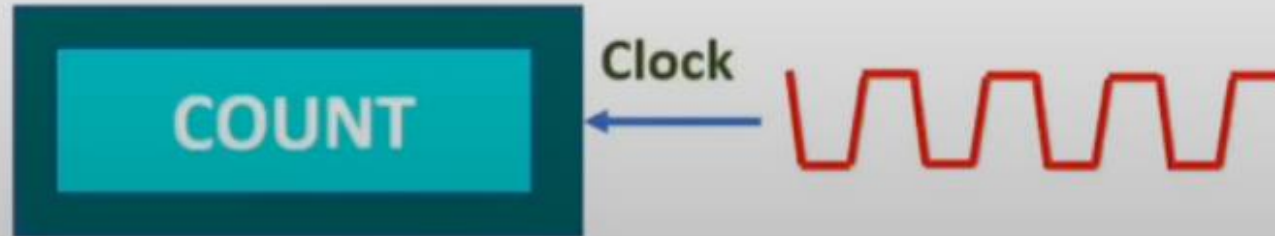
- ❑ This Timers are Up counter.
- ❑ So, on given clock it will increment by 1.
- ❑ When it reaches to FFFFH, it will rolls back to 0000H and during that it will generates Timer Overflow interrupt.

$$\text{Count} = \text{FFFFH} - \text{Value} + 1$$

- ❑ So, if you wants to count 9 then $\text{Count} = \text{FFFF} - 9 + 1 = \text{FFF7H}$

MOV TH0, #FFH

MOV TL0, #F7H



- ❑ This count loaded in T0 or T1 will increment after every clock.

☐ **Timer or Counter?**

- ☐ If clock to the count is given by internal clock of 8051 then it will be timer and if clock is given by external clock on T0 and T1 then it will counter.
- ☐ That is to be configured by TMOD register of 8051.
- ☐ T/C bit will decide timer or counter configuration of 8051.

❑ How Timer / Counter works?

Load Count in T0 or T1

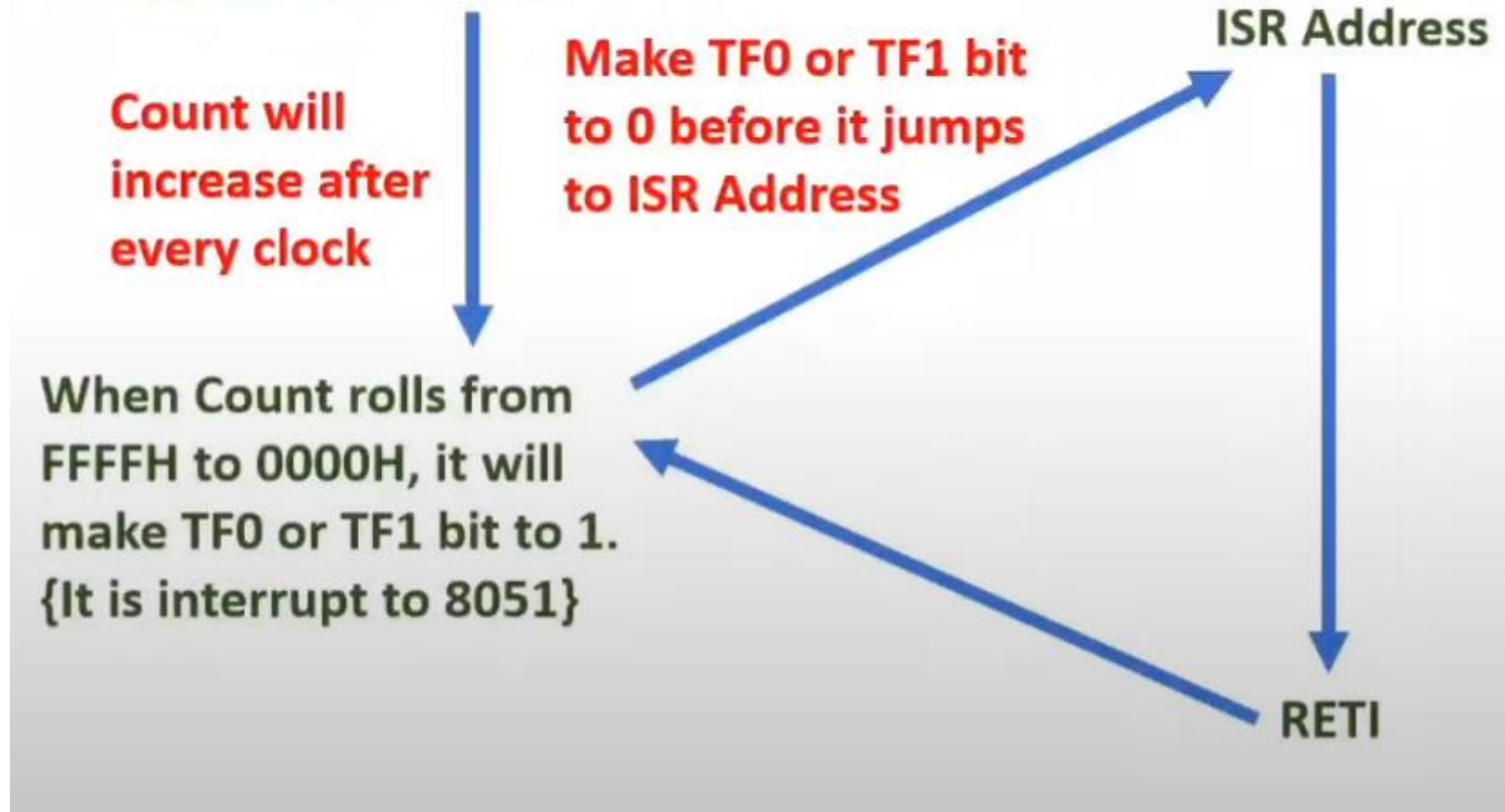
Count will
increase after
every clock

Make TF0 or TF1 bit
to 0 before it jumps
to ISR Address

ISR Address

When Count rolls from
FFFFH to 0000H, it will
make TF0 or TF1 bit to 1.
{It is interrupt to 8051}

RETI



TCON and TMOD Registers

- ☐ 8051 has Two 16 bits Timers T0 & T1, working as up counters. T0 and T1 is further divided into 8bits of registers TH0-TL0 and TH1-TL1.
- ☐ If T0 & T1 counts internal clock pulses, then it is timer.
- ☐ If T0 & T1 counts External clock pulses, then it is Counter.
- ☐ Timer action is controlled by TCON and TMOD registers.
- ✓ TCON register – {Bit Address TCON.7 to TCON.0}

TCON and TMOD Registers

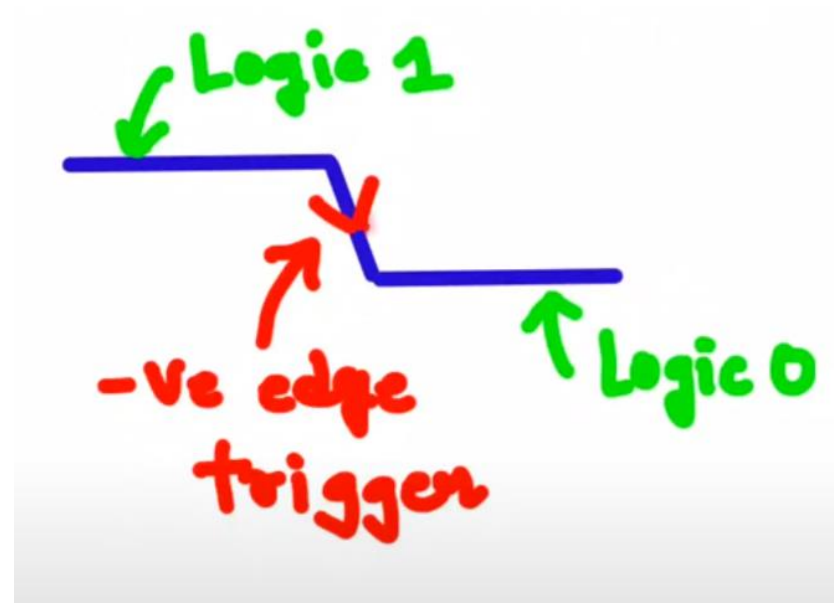
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- ✓ TCON register – {Bit Address TCON.7 to TCON.0}

✓ TCON register – {Bit Address TCON.7 to TCON.0}

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
-----	-----	-----	-----	-----	-----	-----	-----

- ☐ **TF1 and TF0** – Timer Overflow Flag
 - ☐ SET 1 = When timer 1 and timer 0 overflows, when timer roll overs to all 0's.
 - ☐ Clear 0 = When processor executes ISR after overflow. {For Timer 1 ISR address is 001BH and Timer 0 ISR address is 000BH}
- ☐ **TR1 and TR0** – Timer Run Control Bit
 - ☐ SET 1 = Start Counting Timer.
 - ☐ Clear 0 = Halts Timer.
- ☐ **IE1 and IE0** – External Interrupt bit
 - ☐ SET 1 = when 8051 receives interrupt on $\overline{INT1}$ and $\overline{INT0}$.
 - ☐ Clear 0 = when ISR executed. {For $\overline{INT1}$ ISR address is 0013H and $\overline{INT0}$ ISR address is 0003H}

- ❑ **IT1 and IT0** – External Interrupt Type bit
- ❑ SET 1 = $\overline{INT1}$ and $\overline{INT0}$ must be –ve edge trigger.
- ❑ Clear 0 = $\overline{INT1}$ and $\overline{INT0}$ must be low level trigger.



✓ **TMOD register – Timer Mode Control register**



- ☐ **$C/\bar{T}$** – Counter / Timer Type bit
 - ☐ SET 1 = Acts as Counter. {External Frequency on T1 & T0}
 - ☐ Clear 0 = Acts as Timer. {Internal Frequency $F_{osc}/12$ }
- ☐ **GATE** – Gate Enable Control bit
 - ☐ SET 1 = Timer Controlled by Hardware. {INTX Signal}
 - ☐ Clear 0 = Timer independent on INTX signal.
- ☐ **M1 & M0** – Mode Control bits

M1	M0	Timer Mode
0	0	Timer Mode 0
0	1	Timer Mode 1
1	0	Timer Mode 2
1	1	Timer Mode 3

Work with Time/ Counter

✓ TCON register – {Bit Address TCON.7 to TCON.0}

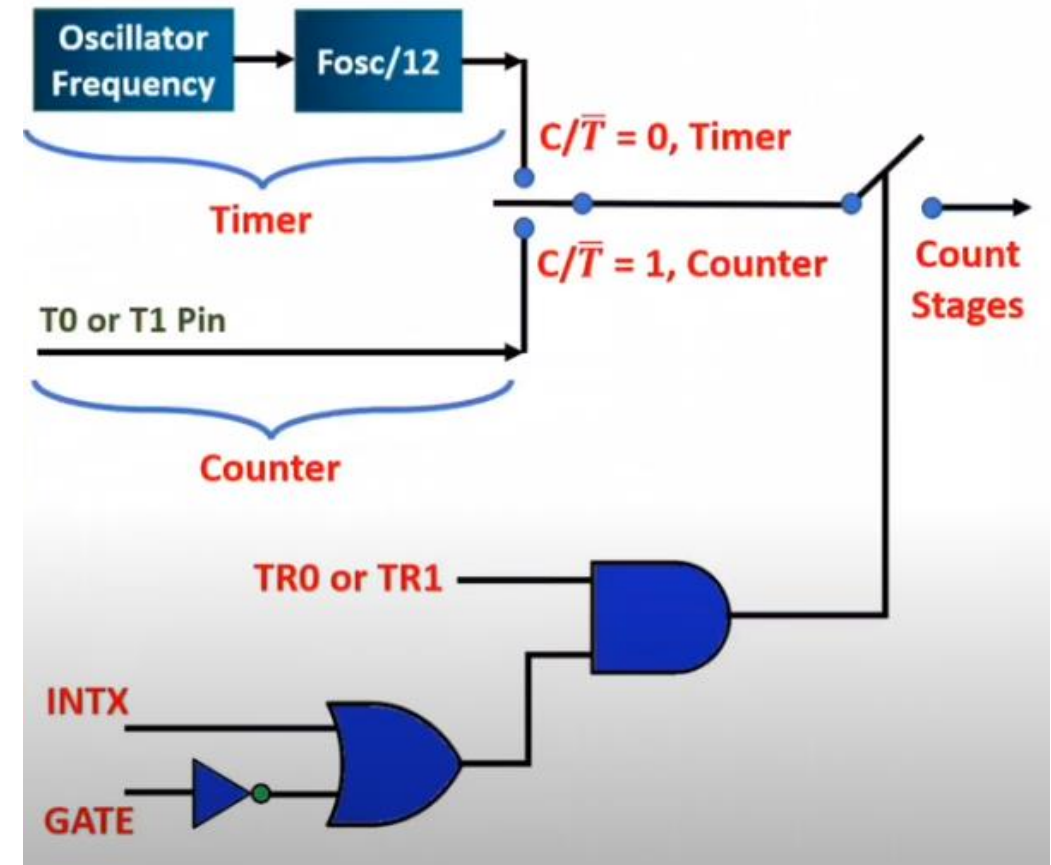
TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
-----	-----	-----	-----	-----	-----	-----	-----

✓ TMOD register – Timer Mode Control register

GATE	$C/\bar{T}$	M1	M0	GATE	$C/\bar{T}$	M1	M0
------	-------------	----	----	------	-------------	----	----



- ❑ 8051 has Two 16 bits Timers T0 & T1, working as up counters.
- ❑ By $C/\bar{T}$ bit we can select timer and counter.
- ❑ In Timer, clock will be given by internal clock.
- ❑ In counter, clock will be given by T0 or T1 Pin of 8051.



Work with Time/ Counter

✓ TCON register – {Bit Address TCON.7 to TCON.0}

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
-----	-----	-----	-----	-----	-----	-----	-----

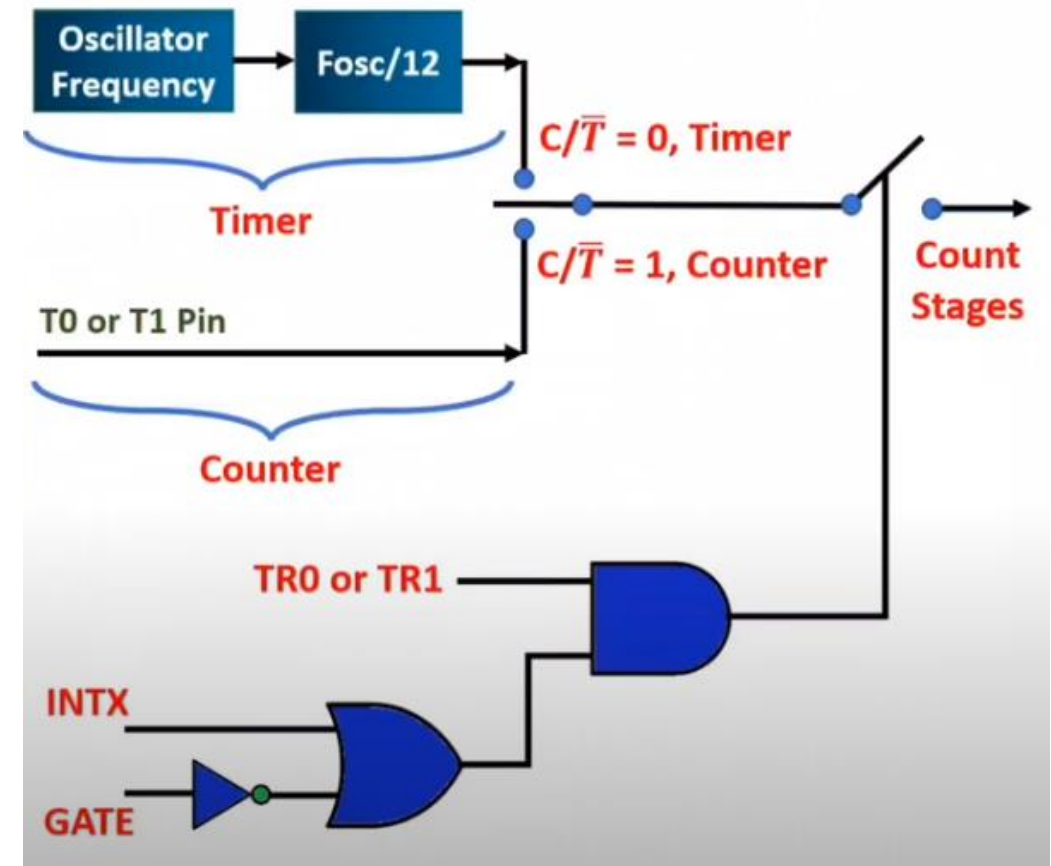
✓ TMOD register – Timer Mode Control register

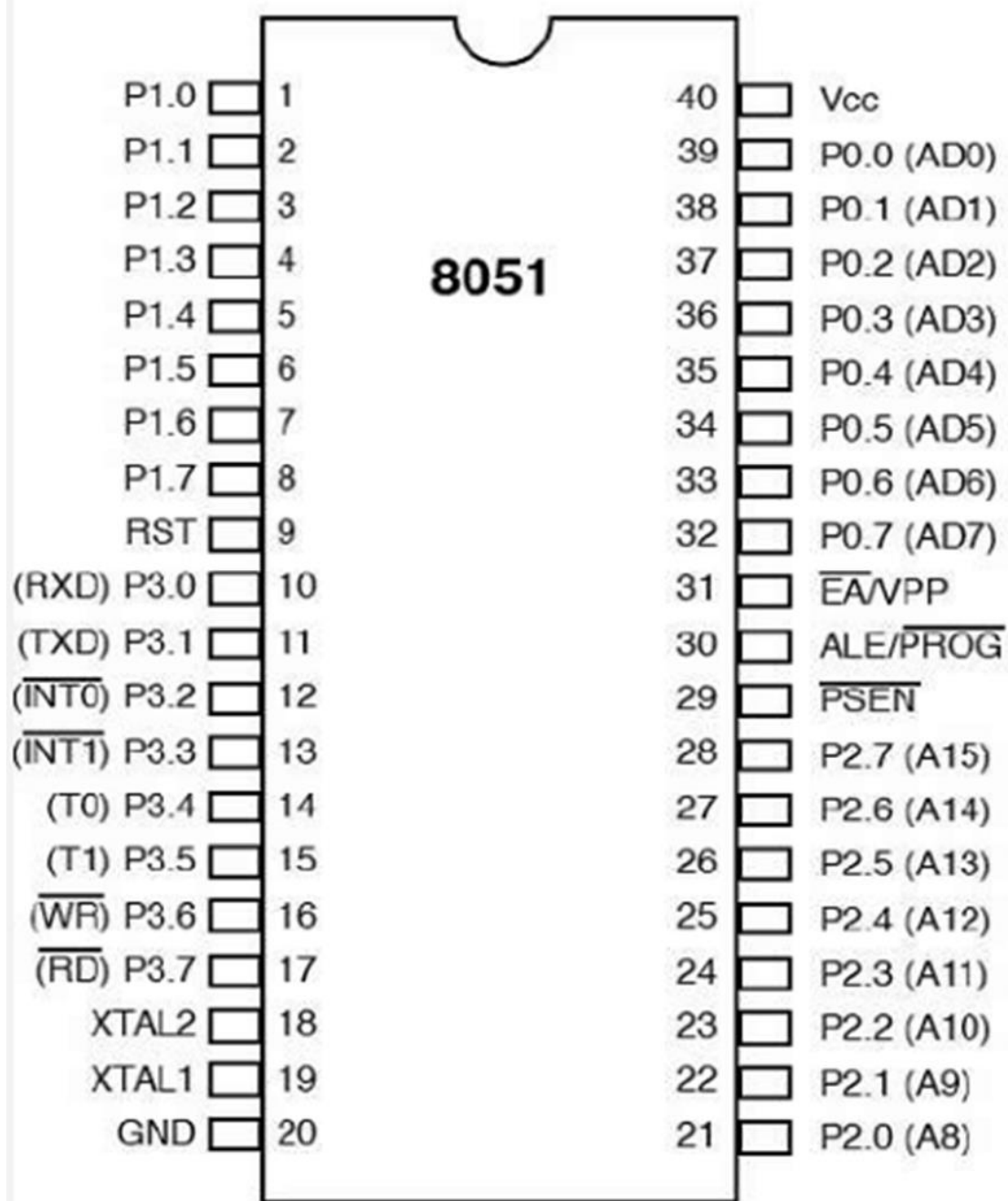
GATE	C/ $\bar{T}$	M1	M0	GATE	C/ $\bar{T}$	M1	M0
------	--------------	----	----	------	--------------	----	----

Timer 1

Timer 0

- ❑ To have running counter, TR bit of TCON register must be 1.
- ❑ If Timer/Counter is triggered by external signal then GATE = 1 of TOMD register, which means Timer/ Counter operation will get trigger by INTX {INT0 or INT1}.
- ❑ If Timer 0 is configured then with GATE bit we use INT0 hardware interrupt to trigger Timer/Counter.
- ❑ If Timer 1 is configured then with GATE bit we use INT1 hardware interrupt to trigger Timer/Counter.
- ❑ If GATE bit is logic 1, then INTX pin will used for timer/Counter only.





Modes of Timer and Counter

✓ Timer Mode 0 {13 bits Timer/ Counter}



- ❑ TLX has 5 bits for count and THX has 8 bits for count. So in total, 13 bits of count is available in this mode 0.
- ❑ After 32 counts TLX rolls over and it will increment THX.
- ❑ So TLX will divide the frequency by 32.
- ❑ By this mode, total maximum count can be $2^{13} = 8K$.
- ❑ So maximum delay = $8192 (12/F_{osc})$

Modes of Timer and Counter

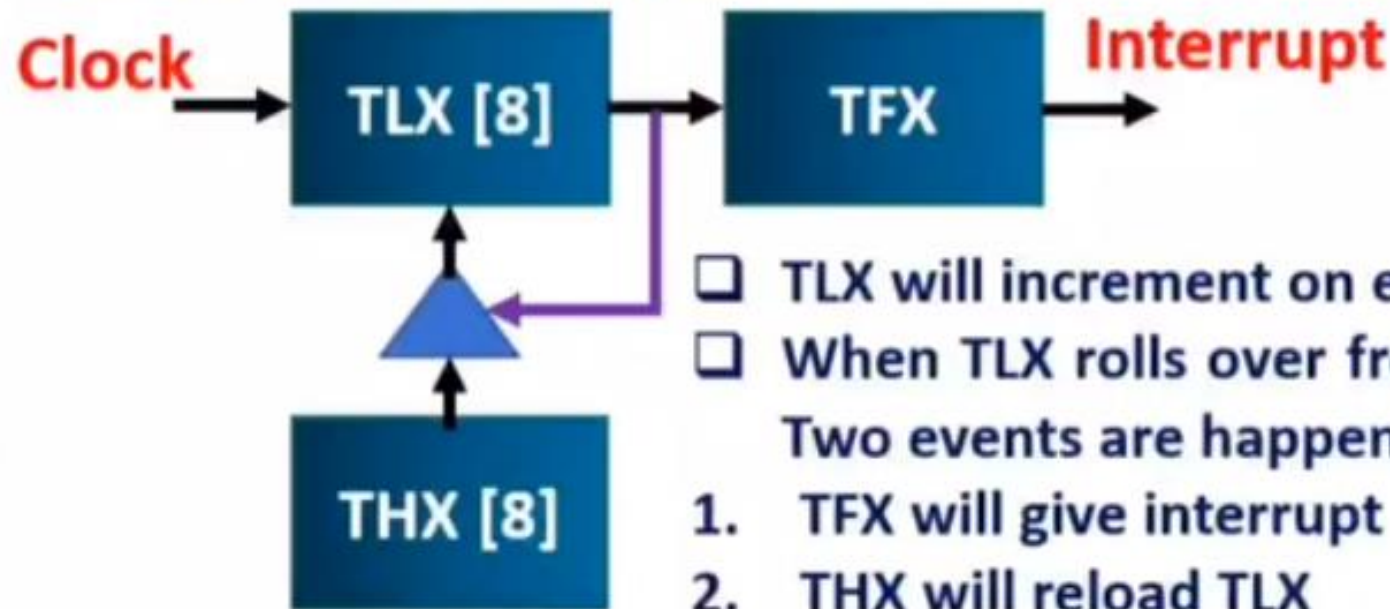
✓ Timer Mode 1 {16 bits Timer/ Counter}



- ☐ TLX and THX used completely here with Mode 1.
- ☐ On each clock 16 bits will increment by 1.
- ☐ TFX will set to 1, when all 16 bits rolls from FFFFH to 0000H.
- ☐ By this mode, total maximum count can be $2^{16} = 64K$.
- ☐ So maximum delay = $65536 (12/F_{osc})$

Modes of Timer and Counter

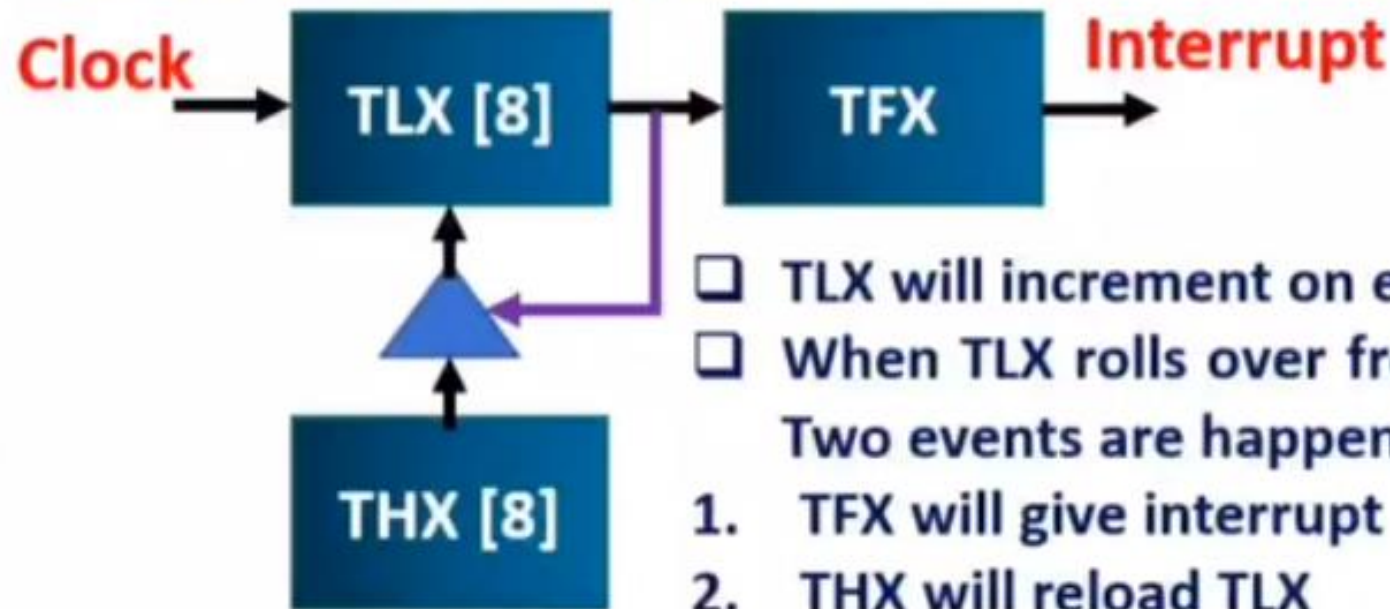
✓ Timer Mode 2 {8 bits Auto reload TL from TH}



- ☐ TLX will increment on every count.
- ☐ When TLX rolls over from FFH to 00H, Two events are happening.
 1. TFX will give interrupt
 2. THX will reload TLX
- ☐ Maximum count = $2^8 = 256$
- ☐ Maximum delay = $256 (12/F_{osc})$

Modes of Timer and Counter

✓ Timer Mode 2 {8 bits Auto reload TL from TH}



- ☐ TLX will increment on every count.
- ☐ When TLX rolls over from FFH to 00H, Two events are happening.
 1. TFX will give interrupt
 2. THX will reload TLX
- ☐ Maximum count = $2^8 = 256$
- ☐ Maximum delay = $256 (12/F_{osc})$

✓ Timer Mode 3 {Two 8 bits timer by Timer 0}



- ☐ TL0 and TH0 used with two separate timers.
- ☐ TL0 will give interrupt to TF0 flag bit of Timer 0.
- ☐ TL0 can be used as Timer and Counter.
- ☐ TH0 will give interrupt to TF1 flag bit of Timer 1.
- ☐ TH0 can only be used as Timer.

Time Programming in 8051

❑ Write a program to Generate delay of 20 μ Sec and send logic 1 on P2.0. Assume Fosc = 12MHz

✓ TCON register – {Bit Address TCON.7 to TCON.0}

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
-----	-----	-----	-----	-----	-----	-----	-----

✓ TMOD register – Timer Mode Control register

GATE	C/ $\bar{T}$	M1	M0	GATE	C/ $\bar{T}$	M1	M0
Timer 1				Timer 0			

- ❑ For Timer 0 with Mode 1 as 16 bits timer, TMOD = 0000 0001B
- ❑ To start Timer 0 with mode 1, TCON = 0001 0000B
- ❑ To stop time 0 with mode 1, TCON = 0000 0000B

- ❑ To calculate Count, one count time = $12/F_{osc} = 1\mu\text{Sec}$.
- ❑ So value of Count = 20 = 14H
- ❑ As timer is up counter actual value should be loaded will be
Count = FFFFH – 14H + 1 = FFECH
- ❑ TL0 = ECH and TH0 = FFH, to be loaded for delay of 20 μ Sec.

```
MOV TMOD, #00000001B    ;Timer 0 Mode 1
MOV TL0, #ECH             ;Count 20 = 14H
MOV TH0, #FFH             ;Start Timer
MOV TCON, #00010000B     ;wait for 20 $\mu$ Sec
Wait: JNB TCON.5, Wait    ;logic '1' on P2.0
      SETB P2.0
      MOV TCON, #00000000B ;Stop Timer
Here: SJMP Here           ;End of Program
```

Time Programming 2

❑ Write a program to Generate square wave of 1KHz on TxD pin. Assume Fosc = 12MHz

✓ TCON register – {Bit Address TCON.7 to TCON.0}

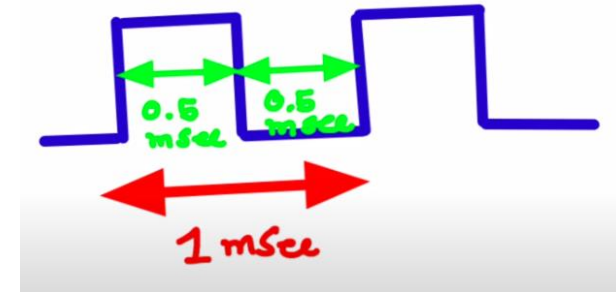
TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
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✓ TMOD register – Timer Mode Control register

GATE	C/ $\bar{T}$	M1	M0	GATE	C/ $\bar{T}$	M1	M0
Timer 1				Timer 0			

- ❑ For Timer 0 with Mode 1 as 16 bits timer, TMOD = 0000 0001B
- ❑ To start Timer 0 with mode 1, TCON = 0001 0000B
- ❑ To stop time 0 with mode 1, TCON = 0000 0000B

- ❑ Square wave of 1KHz has time = 1msec.
- ❑ So for 0.5msec, It should be high and for 0.5msec, it should be low.
- ❑ To calculate Count, one count time = $12/F_{osc} = 1\mu\text{Sec}$.
- ❑ So value of Count = $0.5\text{msec}/1\mu\text{Sec} = 500 = 1F4H$
- ❑ As timer is up counter actual value should be loaded will be
Count = $FFFFH - 1F4H + 1 = FE0CH$
- ❑ TL0 = 0CH and TH0 = FEH



```

Repeat: CLR P3.1                ;Clear TxD line
        MOV TMOD, #00000001B    ;Timer 0 Mode 1
        MOV TL0, #0CH
        MOV TH0, #FEH           ;Count 500 = 1F4H
        MOV TCON, #00010000B    ;Start Timer
Wait:   JNB TCON.5, Wait         ;wait for 0.5mSec
        CPL P3.1                ;Square wave
        MOV TCON, #00000000B    ;Stop Timer
        SJMP Repeat             ;Repeat of Program
    
```